Manuscript template (IMRaD)

Original research article, section by section

Foundations of Clinical Research · Sessions 7–8 (7.3, 8.1 Writing the paper) · Template version 1.0

How to use this template

- Replace every [grey italic text in brackets] with your own text. Press Ctrl + H and search for "[" to step through them, and check none remain before you share the document.
- Grey italic guidance and shaded "Example" boxes are there to help you. Delete them as you write: Home → Styles → right-click "Guidance" (or "Example") → Select All → Delete.
- Write in this order: tables and figures, Methods, Results, Introduction, Discussion, Abstract, Title. Then reformat to your target journal's instructions for authors (word limits, reference style, headings). They always win over this template.
- Remove the Neudata header and this guidance before submission. Most journals want double line spacing and line numbers (Layout → Line Numbers → Continuous).
- **In the course:** each participant uses this template for their own draft manuscript on one of the five FLUID-ICU questions (Q1 to Q5): a STROBE-guided original article of about 2,000 words, written from the results pack for that question in Course/Data/results_packs/. Methods and Results are modelled in Session 7, the rest in Session 8, and you write your own as homework. See Assignment/manuscript_brief.md.

Reporting guideline reminder

- **CONSORT**: randomised trials (with the CONSORT flow diagram).
- **STROBE**: observational studies (cohort, case-control, cross-sectional).
- **STARD**: diagnostic accuracy studies.
- **PRISMA**: systematic reviews and meta-analyses (with the PRISMA flow diagram).
- Others (SPIRIT protocols, CARE case reports, SQUIRE quality improvement, TRIPOD+AI prediction models, SRQR/COREQ qualitative): find them all at **www.equator-network.org**. Fill in the checklist with page numbers and submit it with the paper.

Title page

Title:	*[Population, intervention/exposure, outcome and design: "...: a randomised controlled trial"]*
Running title:	*[≤ 50 characters]*
Authors:	*[First Last]*[1], *[First Last]*[2] ... (ORCID iDs)
Affiliations:	*[[1] Department, Institution, City, Country] [[2] Department, Institution]*
Corresponding author:	*[Name, postal address, email, phone, ORCID]*
Word counts:	*[Abstract n] · [Main text n] · [n tables, n figures]*
Article type:	*[Original research]*

Example: *Restrictive versus usual fluid strategy and 28-day mortality in adults with septic shock: a pilot randomised controlled trial.*

Abstract

Structured, usually ≤ 250 words. It must stand alone: the numbers in it must match the Results exactly. No spin: report the primary outcome first, with its effect size and 95% CI, even if it is not significant.

Background: *[Problem, gap, aim (2–3 sentences)]*

Methods: *[Design, setting, participants, intervention/exposure, primary outcome, analysis]*

Results: *[n analysed; primary outcome per group; effect size (95% CI); key secondary outcomes; harms]*

Conclusions: *[Answer the question, in words that match the design]*

Trial registration: *[Registry and number (trials)]*

Keywords: *[3–6 keywords, ideally MeSH terms, not repeating the title]*

> **Example:** *Septic shock; Fluid therapy; Water-electrolyte balance; Intensive care units; Mortality*

Introduction

A funnel of three short paragraphs (about 400–500 words), from broad to your question.

Paragraph	What it does	Example sentence
1. What is known	The problem and its size; what is established (cited)	*Septic shock is common in ICUs and fluid is a core part of its treatment [1].*
2. What is unknown	The gap, conflict or limitation in the evidence	*Observational studies link a positive fluid balance to death [2], but cannot separate the effect of fluid from severity; trials are few and small [3].*
3. What we did	The aim or hypothesis, in one or two sentences	*We therefore conducted a pilot randomised trial to compare ...*

[Write the Introduction here]

Methods

Enough detail for another team to repeat the study. Past tense. Follow your reporting guideline item by item.

Study design and setting

Design, where, when (dates), registration number; protocol availability.

[Write here]

Participants

Eligibility criteria; how participants were identified and recruited.

[Write here]

Intervention / exposure and comparator

What exactly was done or measured, by whom, for how long.

[Write here]

Outcomes

Primary outcome (only one) and secondary outcomes, with definitions and time points.

[Write here]

Data collection

Sources, CRF, data management and quality checks.

[Write here]

Sample size

The calculation and all its assumptions.

[Write here]

Statistical analysis

Descriptive statistics; primary analysis with effect measure and 95% CI; adjustment; secondary and subgroup analyses; missing data; software; significance level. State that a SAP was written in advance.

[Write here]

Ethics

Committee name and approval number; consent procedure (or waiver).

[Write here]

Results

Participants FIRST: the flow diagram (screened, eligible, included, analysed, with reasons at each step) and Table 1. Then the primary outcome, then secondary outcomes and harms. Numbers with effect sizes and 95% CIs; no interpretation; each number in one place only (text OR table, refer to tables rather than repeating them).

Participants

[Flow (Figure 1) and baseline characteristics (Table 1)]

Primary outcome

[Result per group, effect size (95% CI), p]

Secondary outcomes

[Results (Table 2)]

Harms / adverse events

[Results]

> **Example:** *Of 412 patients screened, 60 were randomised and 58 analysed (Figure 1). By day 28, 9 of 29 (31%) patients in the restrictive group and 11 of 29 (38%) in the usual-care group had died (risk ratio 0.82, 95% CI 0.40 to 1.68). [Fictional numbers, for illustration only.]*

Discussion

Six moves, in this order. Calibrate your language to the design: "was associated with" for observational studies; "caused" only for well-conducted trials; never "a trend" for p = 0.42.

Move	What to write
1. Answer	The main finding in one or two sentences, answering the question from the Introduction.
2. Compare	How it agrees or disagrees with previous studies (cited).
3. Explain	Possible explanations and mechanisms.
4. Limitations	Strengths and limitations, honestly: bias, confounding, precision, generalisability and how each might change the result.
5. Implications	For practice and for future research (what study is needed next).
6. Conclude	A closing sentence that matches the design and the data.

[Write the Discussion here]

Conclusion

Two or three sentences. No new results; no claims beyond the data.

> **Example:** *In this pilot trial, a restrictive fluid strategy was feasible. Whether it reduces mortality needs a larger trial.*

[Conclusion]

Declarations

Required by almost every journal. Check the journal's exact headings.

Declaration	Text
Ethics approval	*[Ethics committee name]*, approval no. *[number]*, dated *[DD/MM/YYYY]*
Consent	*[Written informed consent / representative / deferred consent / waiver granted by ...]*
Registration	*[Registry, number, date (prospective / retrospective)]*
Funding	*[Funder, grant number, role of the funder, or "No specific funding"]*
Competing interests	*["The authors declare no competing interests" or details (ICMJE form)]*
Data availability	*[Where and how the de-identified data and code can be obtained]*
Use of AI	*[Which AI tools were used, for what (e.g. language editing), and that the authors take responsibility for the content, or "No AI tools were used"]*
Acknowledgements	*[Contributors who do not meet authorship criteria]*

Author contributions (CRediT)

Tick each role for each author (initials as column headings).

CRediT role	[AB]	[CD]	[EF]	[GH]	[IJ]
Conceptualization	☐	☐	☐	☐	☐
Data curation	☐	☐	☐	☐	☐
Formal analysis	☐	☐	☐	☐	☐
Funding acquisition	☐	☐	☐	☐	☐
Investigation	☐	☐	☐	☐	☐
Methodology	☐	☐	☐	☐	☐
Project administration	☐	☐	☐	☐	☐
Resources	☐	☐	☐	☐	☐
Software	☐	☐	☐	☐	☐
Supervision	☐	☐	☐	☐	☐
Validation	☐	☐	☐	☐	☐
Visualization	☐	☐	☐	☐	☐
Writing – original draft	☐	☐	☐	☐	☐
Writing – review & editing	☐	☐	☐	☐	☐

References

Use the journal's style (often Vancouver: numbered in order of citation). Manage them with a reference manager. Cite original studies, not only reviews; check every reference against the source.

1. *[Authors. Title. Journal. Year;Volume:Pages. doi]*
2. *[Authors. Title. Journal. Year;Volume:Pages. doi]*
3. *[Authors. Title. Journal. Year;Volume:Pages. doi]*

Tables and figures

Each on its own page after the references (unless the journal says otherwise). Every table and figure must stand alone: a clear title, units, n per column, abbreviations explained in a footnote.

Table 1. Baseline characteristics of participants

Characteristic	Group A (n = __)	Group B (n = __)
[Characteristic, unit, summary]		
[Characteristic, unit, summary]		
[Characteristic, unit, summary]		

Data are n (%), mean (SD) or median (IQR). Abbreviations: ...

Table 2. Primary and secondary outcomes

Outcome	Group A	Group B	Effect (95% CI)	p
[Primary outcome]				
[Secondary outcome]				

Figure 1. Participant flow diagram

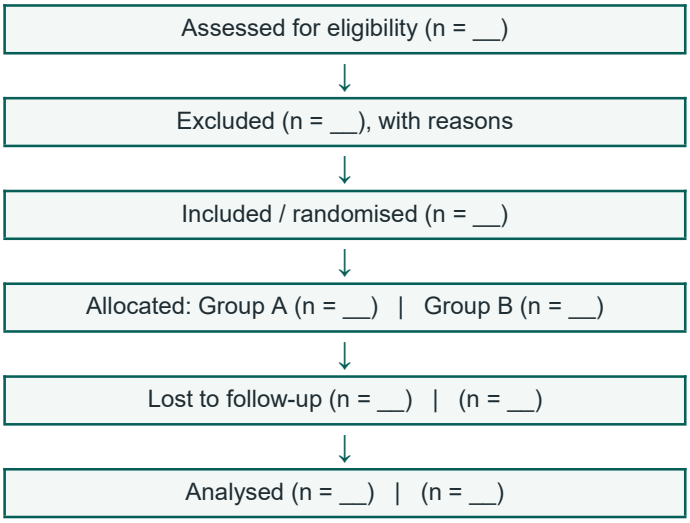


Figure legends: one per figure, giving the title, what is shown and abbreviations.